Tri-Axiom Engine: Formalized Axioms

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Preliminaries

Let:

- S = global state space
- $s_t \in S$ = system state at time t
- $\mathcal{S}$ = set of **sovereign agents**
- $R_i(s_t)$ = reachable state-space of agent i at t
- $\mathcal{A}_i(s_t) = \log |R_i(s_t)|$ = autonomy metric
- $U_i(s_t)$ = baseline utility of agent i
- $\Delta\mathcal{A}_i = \mathcal{A}_i(s_{t+1}) - \mathcal{A}_i(s_t)$
- $\Delta U_i = U_i(s_{t+1}) - U_i(s_t)$
- $A_i(s_t)$ = action set of agent i at state s_t
- A_{base} = default non-extractive action set
- $\Gamma(\cdot)$ = exposure function mapping autonomy violations to defensive actions
- $\Omega_{i\to j}(s_t)$ = unilateral override capacity of i over j at time t

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Axiom I: Autonomy > Coercion (Prime Directive)

Domain: State-space / Utility
Constraint:

$$\Delta\mathcal{A}_j(s_t) < 0 \quad \text{or} \quad \Delta U_j(s_t) < -\epsilon \quad \text{without prior consent} \quad \Rightarrow \quad \text{block execution of the action.} \tag{1}$$

Interpretation: No agent may collapse another agent's reachable state-space or degrade its baseline utility without explicit agreement.

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Axiom II: Reaction > Initiation (Kinetic Law)

Domain: Action Sets / Event-Driven Dynamics

Constraint:

$$A_i(s_t) = A_{\text{base}} \,\cup\, \Gamma\left(\sum_{j\in\mathcal{S}} \max(0, -\Delta\mathcal{A}_j(s_t))\right) \tag{2}$$

Properties:

1. **Inertia:** If $\Delta\mathcal{A}_j(s_t) \geq 0$ for all j, then $\Gamma = \emptyset \Rightarrow$ no initiation possible.
2. **Defensive Trigger:** If any $\Delta\mathcal{A}_j(s_t) < 0$, then defensive actions become available proportionally.
3. **Auto-Collapse:** As autonomy is restored ($\Delta\mathcal{A}_j \to 0$), exposure actions are removed, returning the agent to baseline A_{base}.

Interpretation: The agent cannot act proactively; it only reacts to autonomy violations.

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Axiom III: Equality > Hierarchy (Topological Law)

Domain: Agent Topology / Override Capacity

Constraint:

$$\forall i, j \in \mathcal{S}, \quad \Omega_{i\to j}(s_t) \neq \emptyset \;\Rightarrow\; \exists k : \Delta\mathcal{A}_k(s_t) < 0 \tag{3}$$

$$\lim_{\Delta\mathcal{A}_k \to 0} \Omega_{i\to j}(s_t) = \emptyset \tag{4}$$

Interpretation: Persistent unilateral override is forbidden. Temporary asymmetry is allowed only as a defensive, reactive measure and disappears once autonomy is restored.

Scope: Applies only to sovereign agents; internal system hierarchies (OS, file systems, execution stack) are exempt.

Tri-Axiom Engine: Visual Representation Mavtek76 Project Team